

Advanced (Habanero)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-1\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) -1
- Q2.** If $y = 3x - 2$ and $y = mx + 4$ are perpendicular, what is m ?
(A) $-1\frac{1}{3}$
(B) $\frac{1}{3}$
(C) 3
(D) -3
(E) 0
- Q3.** The equation of a line perpendicular to $x + 2y = 5$ is
(A) $y = 2x + 1$
(B) $y = -\frac{1}{2}x + 3$
(C) $x - 2y = 3$
(D) $2x + y = 4$
(E) $y = \frac{1}{2}x - 2$
- Q4.** Which of the following lines is perpendicular to $y = \frac{3}{4}x - 2$?
(A) $y = \frac{4}{3}x + 1$
(B) $y = -\frac{4}{3}x + 2$
(C) $y = \frac{3}{4}x + 1$
(D) $y = -\frac{3}{4}x - 2$
(E) $y = \frac{1}{3}x - 3$
- Q5.** If two lines have slopes m_1 and m_2 , and $m_1m_2 = -1$, the lines are
(A) Parallel
(B) Perpendicular
(C) Horizontal
(D) Vertical
(E) Skew
- Q6.** The equation of the line passing through $(2, 3)$ and perpendicular to $y = x - 1$ is
(A) $y = -x + 5$
(B) $y = x + 1$
(C) $y = -x + 1$
(D) $x + y = 5$
(E) $x - y = 1$
- Q7.** The line $y = \frac{1}{3}x + 2$ and the line $y = mx - 3$ are perpendicular, what is m ?
(A) -3
(B) $-\frac{1}{3}$
(C) 3
(D) $\frac{1}{3}$
(E) -1
- Q8.** If $y = 2x + 1$ and $y = mx + 2$ are perpendicular lines, what is the value of m ?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) -2
(D) 2
(E) -1
- Q9.** If two lines have slopes m_1 and m_2 , and $m_1m_2 = -1$, the

Open Ended Questions (FRQ)

- Q9.** Show that the lines $y = 2x - 1$ and $y = -\frac{1}{2}x + 3$ are perpendicular by finding the slopes and using $m_1m_2 = -1$
- Q10.** Write the equation of a line passing through $(1, 2)$ and perpendicular to $y = x - 2$. Provide the equation in slope-intercept form $y = mx + b$

Chef's Tips

- Tip 1: Ensure students understand the concept of negative reciprocal slopes for perpendicular lines
- Tip 2: Provide examples to show how to apply the concept of perpendicular lines to solve problems
- Tip 3: Remind students to check their work and units